Rubber Processing Machinery

Rubber Vulcanization

Microwave Vulcanization Oven (UHF)

Vulcanization system for curing rubber profiles

JWL-12T-X









Microwave vulcanization oven (UHF) is used for vulcanizing rubber seals in high-speed rail part manufacturing, automotive component manufacturing, and aerospace product manufacturing. It utilizes special techniques such as microwave system, electric heating system, and gas heating system to achieve uniform heating effects on rubber strips inside the heating channel. Hot air is introduced into the channel to ensure that both the interior and exterior of the material reach the desired process temperature, achieving optimal vulcanization results.

Application Materials

EPDM rubber, natural rubber, butyl rubber, etc.

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Features

- Utilizes multi-feed technology for even distribution of microwave energy, resulting in uniform heating and high vulcanization efficiency.
- High-stability microwave power supply with high and low voltage isolation, resistant to interference.
- Equipped with comprehensive protection measures for safe and reliable operation, and convenient operation and maintenance.
- The hot air can be achieved through either electric heating or gas heating, selectable through a switch.

Specifications

Dimensions	L9000×W2500×H2300mm (customizable)
Profile Pass-through Aperture Height	Effective size above the conveyor: 100mm
Conveyor Method	Teflon conveyor belt/roller conveyor
Conveyor Speed	4 ~ 35m/min, variable frequency speed regulation
Speed Control	Proportional regulator, -50% ~ +150%
Microwave Power	Standard 12kw, customizable according to customer needs. Magnetron brands available in Hitachi and Panasonic.
Heating Temperature	Gas heating: max. 400℃, common range: 300℃ ~ 350℃ Electrical heating: max. 300℃, common range: 200℃-280℃
Heating Method	Gas heating/electric heating/ gas or electrical heating. Gas heating is available in German Kromschroeder or Weishaupt burners.
Temperature Deviation Within The Channel	Longitudinal temperature difference is less than 10℃, with an accuracy of ±1℃.
Door Opening	Pneumatic or manual opening
Material Detection	Roller detection
Microwave Leakage	At a distance of 5cm from the device, leakage is below 5mW/cm2.
Infrared Temperature Measurement System	Optional feature for surface temperature measurement of rubber strips
Flame Detection and CO2 Injection	Optional feature for flame detection and CO2 injection after detection.
Smoke Exhaust Hood	Optional feature for collecting oil smoke
Electric/Pneumatic Baffle	Optional feature that automatically closes when a flame signal is detected to prevent spread of fire.

Highlights



Microwave Incidence and Reflection Control System (Optional)

We've developed proprietary software that monitors and adjusts both the incoming and reflected microwaves within each microwave unit of the vulcanization oven. When the product load inside the chamber presents low impedance and causes the reflection coefficient to rise above 20%, the system automatically reduces the microwave output power. This dynamic adjustment helps optimize energy efficiency, ensuring maximum utilization of the emitted microwaves while lowering energy consumption and improving overall production performance.

Related Products



Hot Air Vulcanization Oven (HAV)



NIR Unit/Burning Oven



Hot Air Shock Channel



Microwave Vulcanization Oven (UHF)

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Why Choose Us

CAPABILITY

Our capabilities make us a reliable source to satisfy your requirements

CASE STUDIES

We have partnered with brands to develop solutions tailored to their production needs

Complete Lines

for the production of rubber seals and profiles

As a professional manufacturer of rubber equipment, Julong not only offers a wide range of individual machines but also provides customized production lines to meet various production needs for rubber seals and profiles.

Rubber Mono Extrusion Line | Rubber Co-Extrusion Line | Electrostatic Flocking Line | Rubber Coating Line |



COMPLETE LINES

Electrostatic Flocking Line

Julong provides professional electrostatic flocking lines for the flocking process on rubber, which is suitable for customers who require a high production speed, advanced technology, and excellent material quality. The production line can l options according to customer needs.

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Explore Extensive Range of **Rubber Processing Equipment**

Rubber Seal Carrier Handling



Vertical Accumulator

Vertical accumulator is used for storing metal carrier in the co-extrusion lines to ensure uninterrupted production during rubber seal carrier welding.

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Pre-Forming Machine

Pre-forming machine is a high-precision and high-configured metal carrier processing equipment commonly used for the initial shaping of metal carriers.

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Rubber Extrusion



Rubber Extruder

Rubber extruder is applicable for the extrusion of rubber products, including rubber hoses, rubber strips, waterstops, and other similar products. It is used for both single extrusion and co-extrusion molding of rubber.

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Rubber Feeder

Rubber feeder is commonly employed in the rubber manufacturing and cable manufacturing industries for transporting rubber strips to the extruder.

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Flocking Equipment



Flock Recycling Machine

Flock recycling machine is utilized for the recovery of flock fibers after the flocking process. It can be matched with a cartridge-type dust collector to collect the flock fibers effectively.

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Flock Cleaning Machine

Flock cleaning machine is used to clean the surface of flocked seals after curing.

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INDIVIDUAL MACHINES

- Rubber Seal Carrier Handling
- Extrusion
- Rubber Vulcanization
- Flocking
- Complementary
- Microwave Thawing Machine
- Other Microwave Systems

COMPLETE LINES

- Rubber Mono Extrusion Line
- Rubber Co-Extrusion Line
- Electrostatic Flocking Line
- Rubber Coating Line

INDUSTRIES

- Automotive
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Meat

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